

SABANCI UNIVERSITY

Quantitative Analysis of Isotretinoin Toxicity and Patient Sentiment

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1. INTRODUCTION AND MOTIVATION

Isotretinoin is widely regarded as the most effective clinical treatment for severe, recalcitrant acne; however, it carries a significant and often unpredictable burden of physiological and psychological side effects. Historically, medical datasets examine these side effects purely through clinical trials, often neglecting the organic, psychological experiences of the patients themselves during the course of the therapy.

As a student currently undergoing this treatment, my primary motivation for this project is deeply personal and analytically driven. I aim to apply rigorous data science methodologies to determine if there is a statistical causation or correlation between daily prescribed dosage levels, the objective severity of reported side effects, and the subjective sentiment of the patient. Ultimately, this project transitions from descriptive analysis into predictive analytics. By developing machine learning models, the goal is to create an intelligent framework that can anticipate toxicity risk and specific adverse events (such as depression) based on patient demographics and dosage, thereby providing a data-driven tool for better symptom management and clinical oversight.

2. DATA SOURCE AND COLLECTION METHODOLOGY

To ensure a robust analysis and meet the course's data enrichment requirement, this research integrates two distinct secondary datasets, effectively combining objective clinical records with subjective patient narratives.

2.1 Primary Features and Sources

- **FDA Adverse Event Reporting System (FAERS):** The clinical foundation of the project was extracted via a Python API query to the official FDA database. The initial pull yielded 3,000 raw clinical records. After rigorous cleaning to remove missing values and inconsistencies, the refined dataset consisted of over 1,200 pristine records featuring patient demographics (Age, Gender), prescribed dosage (mg), and specific recorded adverse reactions.
- **Drugs.com Patient Reviews:** Sourced from a publicly available dataset on Kaggle, this dataset contains unstructured textual reviews written by individuals undergoing treatment. The data was programmatically filtered to isolate records specifically related to the active ingredient 'Isotretinoin'.

2.2 Data Enrichment (Feature Engineering and NLP)

To bridge the operational gap between clinical data and text reviews, significant data enrichment was performed. First, a quantitative `severity_score` was engineered in the FDA dataset by algorithmically counting the distinct number of side effects reported per patient profile. Second, Natural Language Processing (NLP) techniques, specifically advanced Regular Expression (Regex) pattern matching, were utilized to extract numerical daily dosages directly from the unstructured patient review

texts. This enrichment pipeline successfully normalized data points across both datasets, allowing for cross-dataset hypothesis testing regarding dosage impacts.

3. DATA ANALYSIS TECHNIQUES

The analysis was conducted through a rigorous multi-stage pipeline, ensuring statistical validity and structural independence before applying predictive modeling algorithms.

3.1 Stage I: Preprocessing and EDA

Exploratory Data Analysis (EDA) focused on understanding demographic characteristics and treatment distributions. Visualizations revealed that patient dosages are heavily clustered around standard clinical prescriptions (20mg, 40mg, and 60mg daily). Furthermore, the demographic distribution confirmed that the patient age heavily skews towards the 15-25 age group, aligning perfectly with dermatological and epidemiological expectations for severe acne incidence.

3.2 Stage II: Inferential Statistics (Hypothesis Testing)

To formally evaluate the core assumptions underlying the treatment experience, three primary hypotheses were statistically tested:

- **H1 (Dosage vs. Severity):** A correlation test revealed no strong linear relationship ($p > 0.05$) between raw daily dosage and the computed side effect severity score, suggesting that physiological toxicity is highly idiosyncratic rather than linearly dependent on pill size.
- **H2 (Dosage vs. Sentiment):** Regression analysis on the review dataset showed no negative correlation between dosage levels and patient satisfaction. Ratings remained consistently high (concentrated between 8-10/10) even at maximum dosages (80mg+), indicating that the drug's therapeutic efficacy overwhelmingly outweighs its physiological hardships.
- **H3 (Gender vs. Severity):** An Independent T-Test was conducted to evaluate treatment tolerance between biological sexes. The test yielded a statistically significant result ($p = 0.004$), proving that male patients report a significantly higher average number of distinct side effects (3.34) compared to female patients (2.75).

3.3 Stage III: Machine Learning Implementation

Three distinct machine learning models were developed to address specific clinical objectives:

- **Model 1 (General Toxicity Risk):** A class-weight balanced Random Forest model was deployed to predict "High Risk" (≥ 2 side effects) vs. "Low Risk" (0-1 side effects) categories based on Age, Gender, and Dosage.
- **Model 2 (Depression Warning System):** A specialized Random Forest classifier aimed at predicting specific, severe psychological adverse events.

- **Model 3 (Sentiment NLP Model):** A Logistic Regression classifier utilizing Term Frequency-Inverse Document Frequency (TF-IDF) vectorization to predict patient satisfaction directly from raw review texts.

4. KEY FINDINGS AND RESULTS

The convergence of exploratory statistics and predictive machine learning models yielded profound insights into medical data modeling and patient psychology:

4.1 Unpredictable Reactions and Clinical Safety (Validating H1)

The lack of linear correlation discovered in H1 was mirrored in the Machine Learning phase. The balanced Random Forest model (Model 1) achieved a modest baseline accuracy of 55.36%, mathematically confirming that idiosyncratic drug reactions are highly complex, non-linear, and difficult to map using basic demographic features alone. However, in Medical AI, it is clinically and ethically safer to over-warn a patient (False Positive) than to miss a severe psychological breakdown (False Negative). The Depression Warning System (Model 2) successfully demonstrated this ethical algorithmic behavior by prioritizing the *Recall* metric (achieving 0.55 for the minority class) over generic accuracy, effectively flagging over half of the high-risk individuals for closer medical monitoring.

4.2 The "Ambivalence" of Medical Text Mining (Validating H2)

The NLP sentiment model (Model 3) achieved an accuracy of 64.71% with a striking 1.00 Recall for Positive reviews, but only 0.08 for Negative reviews. This disparity revealed a fascinating psychological phenomenon: patients reporting horrific physical side effects frequently utilized highly positive "cure" vocabulary (e.g., "miracle," "cleared my acne," "life-changer") within the same text. This linguistic ambivalence makes text-based AI classification incredibly challenging, yet it perfectly validates the H2 statistical finding that the drug's overarching efficacy outweighs its toxicity, maintaining high overall patient satisfaction.

4.3 Biology Trumps Pill Size (Validating H3)

Feature Importance analysis of the predictive models mathematically validated the H3 statistical test. The Random Forest algorithm relied far more heavily on *Gender* than on *Daily Dosage* to make its toxicity predictions. This conclusively proves that treatment tolerance is largely dictated by inherent biological traits and sex-specific baselines rather than a uniform prescription size.

5. LIMITATIONS AND FUTURE WORK

5.1 Limitations

- **Missing Confounding Variables (Weight & Time):** In clinical dermatology, Isotretinoin is prescribed strictly based on body weight (mg/kg), and cumulative dose tracking over 6 months

determines the ultimate success. Because exact patient weight and the specific month of the treatment timeline are omitted in public FAERS data due to privacy constraints, raw daily dosage lacks full context.

- **Self-Reporting Bias:** Both the FDA FAERS database and Drugs.com reviews rely on patient or practitioner self-reporting. This inherently introduces a polarization bias, as individuals are statistically more likely to report extreme experiences (either miraculous cures or severe adverse reactions) while moderate experiences remain underrepresented.
- **Lack of Medical History (Comorbidities):** The datasets do not provide comprehensive pre-existing medical records. It is challenging to definitively isolate whether a side effect like depression was purely drug-induced or an exacerbation of an underlying pre-existing psychological condition.

5.2 Future Extensions

- **Cumulative Dosage Tracking:** Future iterations should incorporate longitudinal clinical APIs to calculate the total cumulative dose reached over months, providing a much stronger indicator of clinical outcomes than static daily dosage.
- **Deep Learning for Text:** Moving beyond traditional TF-IDF and Logistic Regression to medical-specific Large Language Model architectures, such as **BioBERT**, would better capture the complex contextual nuances of ambivalent clinical narratives.
- **Genetics and Lab Results:** To transition this analytical model into a clinical-grade diagnostic tool, future studies must integrate longitudinal patient lab results (e.g., monthly liver enzyme and lipid panels) alongside genetic predispositions.

6. AI ASSISTANCE DISCLOSURE

In accordance with the academic integrity guidelines of Sabancı University, AI tools (LLMs) were utilized strictly for assisting with Regex pattern formulation for text extraction, debugging minor code errors, formatting the document layout, and proofreading the final report. The conceptualization, data analysis, feature engineering, and fundamental interpretations remain entirely my own original work.